

**A PROJECT REPORT
On**

“Hybrid 3D Reconstruction”

**Submitted to
KIIT Deemed to be University**

In Partial Fulfillment of the Requirement for the Award of

**BACHELOR’S DEGREE IN
Computer Science Engineering**

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CERTIFICATE

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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Computer Science & Engineering) at KIIT Deemed to be university, Bhubaneswar. This work is done during year 2025-26, under our guidance.

Date: 02/04/26

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ABSTRACT

This project presents the development of a lightweight and efficient 3D reconstruction pipeline that combines classical geometric principles with modern deep learning techniques. Traditional 3D reconstruction methods, such as voxel carving and graph-cut optimization, require multiple calibrated images and controlled environments, making them computationally expensive and less practical for general use.

The proposed system adopts a depth-based reconstruction approach, where monocular depth estimation serves as the core component. Using the MiDaS (Mixed Depth Scale) model, the pipeline generates dense depth maps from single RGB images without requiring stereo setups or specialized hardware. These depth maps are projected into 3D space to form point clouds, which represent the geometric structure of the scene. The point clouds are then converted into continuous surfaces using Poisson Surface Reconstruction, resulting in smooth and visually coherent 3D meshes.

To improve reconstruction quality, the system incorporates a pseudo-multiview mechanism inspired by multi-view stereo concepts. Instead of using multiple physical images, the system generates additional viewpoints by applying image transformations such as rotation. The resulting point clouds are aligned and merged using the Iterative Closest Point (ICP) algorithm to form a unified 3D representation.

The proposed pipeline provides a computationally efficient solution for generating approximate 3D models from single images, making it suitable for lightweight applications on standard hardware.

Keywords: Depth Estimation, Point Cloud, Mesh Reconstruction, Open3D, MiDaS, Computer Vision

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Chapter 1

Introduction

High-quality 3D models are widely used in applications such as gaming, augmented reality (AR), virtual reality (VR), e-commerce, and animation. Traditionally, creating 3D models requires manual effort using specialized software, making the process time-consuming and dependent on skilled designers.

Recent advancements in computer vision and deep learning have enabled automatic 3D reconstruction from 2D images. However, classical approaches such as voxel carving and multi-view stereo require multiple calibrated images and controlled environments, making them computationally expensive and difficult to deploy in real-world scenarios.

To address these limitations, this project proposes a lightweight hybrid 3D reconstruction pipeline that combines classical geometric principles with modern deep learning techniques. The system uses monocular depth estimation to infer scene geometry from a single image and reconstructs a 3D mesh using point cloud processing and surface reconstruction methods.

Additionally, the system introduces a pseudo-multiview approach by generating multiple rotated views of a single image, improving geometric consistency without requiring multiple physical cameras.

The objective of this project is to develop an efficient and accessible 3D reconstruction pipeline that works on standard hardware while producing visually meaningful 3D models.

Chapter 2

Basic Concepts/ Literature Review

2.1 Classical 3D Reconstruction

Traditional 3D reconstruction techniques include methods such as silhouette-based reconstruction, voxel carving, and multi-view stereo. These approaches rely on multiple images captured from different viewpoints along with accurate camera calibration.

Silhouette-based methods construct a visual hull by intersecting object silhouettes from multiple views, while photo-consistency methods refine the model by ensuring consistent appearance across images. Although these methods produce accurate results, they require controlled environments and high computational resources.

2.2 Depth Based Reconstruction

Recent advancements in deep learning have enabled monocular depth estimation, where a neural network predicts depth information from a single RGB image. Models such as MiDaS allow estimation of relative depth without requiring stereo input or camera calibration.

Depth-based reconstruction pipelines use this estimated depth to generate 3D point clouds by projecting pixels into 3D space using camera geometry. These point clouds are then converted into continuous surfaces using algorithms such as Poisson Surface Reconstruction.

2.3 Hybrid Reconstruction Approach

This project adopts a hybrid approach that combines classical geometric principles with modern depth estimation techniques. Instead of relying on multiple physical views, the system simulates multiple viewpoints by applying image transformations such as rotation.

The generated point clouds from these pseudo-views are aligned using the Iterative Closest Point (ICP) algorithm, which improves geometric consistency.

Chapter 3

Problem Statement / Requirement Specifications

3.1 Problem Statement

To develop a 3D reconstruction system that can generate high-fidelity, low-polygon meshes from a single image in sub-second time, while ensuring the textures are free from baked-in lighting and are correctly UV-unwrapped for professional graphics pipelines.

3.2 System Architecture

The System follows a modular pipeline consisting of:

1. Image Input
2. Background Removal
3. Depth Estimation (MiDas)
4. Depth Processing and Normalization
5. Point Cloud Generation
6. Psuedo-Multiview Generation
7. Point Cloud Alignment(ICP)
8. Point Cloud Fusion
9. Mesh Reconstruction (Poisson Surface Reconstrunction)

3.3 System Design

3.3.1 Design Constraint: Python programming environment,PyTorch framework for depth estimation,Open3D library for point cloud and mesh processing,OpenCV for image preprocessing,Limited GPU memory (optional CUDA support)

3.3.2 System Constraint: While this hybrid pipeline offers a resource-efficient alternative to classical 3D reconstruction, it faces several inherent technical constraints. The overall reconstruction accuracy remains highly sensitive to the quality of the initial input images. Furthermore, the monocular depth estimation provided by MiDaS yields relative rather than absolute depth, which can impact the global scale and metric precision of the model.

Chapter 4

Implementation

4.1 Methodology

The proposed system follows a hybrid reconstruction pipeline that integrates deep learning-based depth estimation with classical geometric processing techniques to generate a 3D model from a single input image.

The process begins with the acquisition of an RGB image, which serves as the primary input to the system. To improve reconstruction quality and eliminate background noise, the input image undergoes preprocessing using an automated background removal technique. This ensures that the reconstruction focuses only on the foreground object, reducing ambiguity in depth estimation.

Once the image is preprocessed, monocular depth estimation is performed using the MiDaS model. This deep learning model predicts a dense depth map representing the relative distance of each pixel from the camera. Since the predicted depth is relative in nature, it is further normalized and refined using smoothing techniques such as Gaussian filtering to reduce noise and improve consistency.

The refined depth map is then used to generate a 3D point cloud by projecting each pixel into 3D space using a pinhole camera model. In this process, the spatial coordinates of each point are computed based on depth values and image geometry, while the corresponding RGB values are retained to preserve visual information.

To enhance reconstruction quality without requiring multiple physical images, the system employs a pseudo-multiview approach. The input image is transformed through controlled rotations to simulate different viewpoints of the same object. Each transformed image undergoes the same depth estimation and point cloud generation process, resulting in multiple partial representations of the object.

These point clouds are then aligned using the Iterative Closest Point (ICP) algorithm, which minimizes the distance between corresponding points across different views. This alignment ensures that all point clouds are brought into a common coordinate system, allowing them to be merged into a single, more complete representation.

Finally, the merged point cloud is converted into a continuous surface using Poisson Surface Reconstruction. This method generates a smooth and coherent 3D mesh by fitting an implicit surface to the point cloud data. The resulting mesh provides an approximate geometric representation of the original object and can be visualized using 3D rendering tools.

4.2 Testing OR Verification Plan

Test ID	Test Case Title	Test Condition	System Behavior	Expected Result
T01	Background Removal	Input image contains complex background with multiple objects	The system applies rembg to isolate the main foreground object	Background is removed and only the object is retained clearly
T02	Depth Estimation	Input image is provided to MiDaS model	The model generates a depth map representing relative distances	Depth map correctly highlights closer and farther regions
T03	Depth Processing	Raw depth map contains noise and irregular values	The system normalizes and smooths depth using filtering techniques	Depth map becomes smooth and suitable for 3D projection
T04	Point Cloud Generation	Processed depth map and RGB image are available	Each pixel is projected into 3D space using camera parameters	A structured point cloud representing object shape is generated
T05	Pseudo-Multiview	Input image is rotated to simulate different views	The system creates multiple transformed images from single input	Multiple viewpoints are generated without additional cameras
T06	ICP Alignment	Multiple point clouds from different views are generated	ICP algorithm aligns point clouds into a common coordinate system	Point clouds are properly aligned with minimal mismatch
T07	Point Cloud Fusion	Aligned point clouds are combined	The system merges all points into a unified dataset	A dense and complete point cloud is formed
T08	Mesh Reconstruction	Final point cloud is provided as input	Poisson Surface Reconstruction is applied to generate mesh	A continuous and smooth 3D mesh is produced
T09	Visualization	Mesh is rendered using Open3D viewer	The system displays the reconstructed 3D model interactively	User can view and inspect the generated 3D model

4.3 Result Analysis OR Screenshots



Figure 4.1: Screenshots of 3D-rendered image from 2d images

4.4 Quality Assurance

The system was tested using multiple input images with varying complexity. Each module of the pipeline was verified individually before integration. Performance and output quality were evaluated visually using Open3D visualization tools.

Chapter 5

Standards Adopted

5.1 Design Standards

The system follows modular software design principles, where each stage of the pipeline is implemented as an independent module. This improves maintainability and scalability. The overall design follows a pipeline architecture, commonly used in computer vision systems.

5.2 Coding Standards

1. Clear and descriptive variable naming
2. Modular function design
3. Consistent indentation and formatting
4. Separation of logic into different files (depth, point cloud, mesh)
5. Avoidance of redundant computations

5.3 Testing Standards

IEEE 829-2008: Followed for software test documentation, ensuring every module—from background removal to mesh generation—was validated through specific test IDs.

Unit & Integration Testing: Each stage of the pipeline (MiDaS depth inference, ICP alignment, Poisson reconstruction) was tested individually before being integrated into the full system.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This project presents a hybrid 3D reconstruction pipeline that combines deep learning-based depth estimation with classical geometric processing techniques. The system successfully generates approximate 3D models from single images using a computationally efficient approach.

While the generated models capture the overall structure of objects, limitations such as lack of fine detail and inaccuracies in complex regions remain.

6.2 Future Scope

Future improvements may include:

1. Using real multi-view images instead of synthetic views
2. Improving depth estimation accuracy
3. Incorporating texture mapping techniques
4. Enhancing mesh refinement and smoothing
5. Exploring advanced neural reconstruction models

These improvements can significantly enhance the quality and realism of reconstructed models.

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CONTRIBUTION REPORT:

Hybrid 3D Reconstruction

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Abstract: This project presents a hybrid 3D reconstruction pipeline integrating classical multi-view consistency with modern AI-driven depth estimation. While traditional voxel-based methods require complex, controlled environments, this implementation utilizes the MiDaS model for rapid monocular depth inference. The system processes 2D inputs, projects them into 3D coordinates, and employs Iterative Closest Point (ICP) alignment for point cloud fusion. Final textured meshes are generated via Poisson Surface Reconstruction, providing a lightweight, high-performance alternative to traditional multi-view stereo approaches for practical applications.

Individual contribution and findings: My primary responsibility involved designing and optimizing a "Hybrid Reconstruction Pipeline" that prioritizes speed and hardware accessibility. While academic literature often describes computationally intensive voxel carving or graph-cut optimizations, this implementation focuses on a depth-based approach to reduce memory overhead and processing time.

Individual contribution to project report preparation: Literature Review and parts Methodology

Full Signature of Supervisor:

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Full signature of the student:

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PLAGIARISM REPORT

(This report is mandatory for all the projects and plagiarism must be below 25%)



